

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 8 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTCVSS801D - MAINTENANCE AND REPAIR OF CONCRETE STRUCTURES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 What is the main cause of corrosion in reinforced concrete structures? a) Low temperature
b) Excessive humidity
c) Chloride ingress and carbonation
d) Use of fine aggregates
- 2 Which metal is most commonly used as reinforcement in concrete structures?
a) Copper
b) Aluminium
c) Steel
d) Lead
- 3 Sulphate attack on concrete primarily affects which component?
a) Coarse aggregate
b) Curing compounds
c) Cement paste
d) Sand particles
- 4 What is a common visual symptom of Alkali-Silica Reaction in concrete?
a) White patches
b) Fine hairline cracks with gel exudation
c) Rust stains
d) Surface scaling
- 5 Which NDT method is used to assess surface hardness of concrete?
a) Ultrasonic Pulse Velocity
b) Core Cutting
c) Rebound Hammer Test
d) Load Testing

- 6 Load testing is used in concrete structures to assess:
- Water absorption
 - Chloride penetration
 - Structural capacity
 - Colour change
- 7 Condition assessment is conducted to:
- Paint the concrete surface
 - Measure sand content
 - Evaluate the extent of damage
 - Estimate cost of materials
- 8 Which is the first step in surface repair strategy?
- Applying sealant
 - Final painting
 - Damage analysis
 - Water curing
- 9 Before applying repair mortar, the surface must be:
- Painted
 - Kept dry
 - Roughened and cleaned
 - Covered with polythene
- 10 Shear strengthening of beams is commonly done by:
- Reducing the beam width
 - Removing cover concrete
 - Adding external stirrups or FRP
 - Using white cement
- 11 Jacketing is a method used for strengthening:
- Beams
 - Slabs
 - Foundations
 - Columns
- 12 Flexural strengthening is mainly intended to improve:
- Fire resistance
 - Load-carrying capacity
 - Colour of concrete
 - Permeability
- Q2. Solve the following.
- What are the reasons for failure of repairs? 6
 - Explain types of rebar 6
- Q3. Solve the following.
- Elaborate the use of chemical admixtures in concrete 6
 - Explain the concept of Supplementary cementitious materials. 6
- Q4. Solve Any Two of the following.
- How to conduct visual inspection in concrete structures? 6
 - Explain the service and exposure conditions of concrete. 6
 - Explain in brief frost attack. Suggest suitable measures to reduce effect of frost attack 6
- Q5. Solve Any Two of the following.

- A) Explain column repair process. 6
- B) Explain the different steps of assessment of condition of RCC structures. 6
- C) What is ground penetrating radar system? How it locate the delaminated concrete? 6

Q6. Solve Any Two of the following.

- A) Explain flexural strengthening methods 6
- B) Explain expansion joint with shear dowels 6
- C) Explain the process of strengthening of slab and beams. 6

*** End ***